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PERFORMANCE SPECIFICATION

PROPELLANTS, DINITROGEN TETROXIDE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

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1. SCOPE

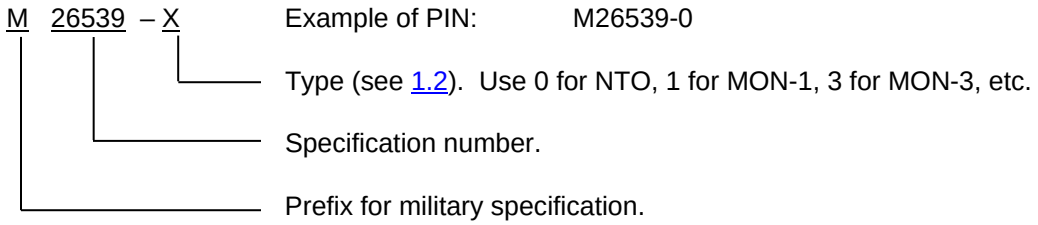
1.1 Scope. This specification covers the requirements for all types and grades of dinitrogen tetroxide (N_2O_4) propellants.

1.2 Classification. Dinitrogen tetroxide propellants will be of the following designated types and MON-3 grades as specified (see 6.2).

Types:

NTO	-	Nominal 99.5% N_2O_4 purity with red-brown color.
MON-1	-	Nominal 99% N_2O_4 and 1% NO with green color.
MON-3	-	Nominal 97% N_2O_4 and 3% NO with green color.
MON-10	-	Nominal 90% N_2O_4 and 10% NO with green color.
MON-15	-	Nominal 85% N_2O_4 and 15% NO with green color.
MON-25	-	Nominal 75% N_2O_4 and 25% NO with green color.

1.3 Part or identifying number (PIN). The PINs to be used for dinitrogen tetroxide acquired to this specification are:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-25508	Propellant, Oxygen

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MIL-PRF-27401	Propellant Pressurizing Agent, Nitrogen
MIL-PRF-27407	Propellant Pressurizing Agent, Helium

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1976	Standard Test Method for Elements in Water by Inductively-Coupled Argon Plasma Atomic Emission Spectroscopy
ASTM E29	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications
ASTM E394	Standard Test Method for Iron in Trace Quantities Using the 1,10-Phenanthroline Method

(Copies of these documents are available from http://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical and physical properties. The chemical and physical properties of the propellant mixture shall conform to the requirements in section 3 and Table I when tested according to the applicable test methods.

3.2 Limiting values. The following applies to all specified limits in this specification: For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off “to the nearest unit” in the last right-hand digit used in expressing the specification limit according to the Rounding Method of ASTM E29.

3.3 Filter. A filter with a rating of at least 10 micrometer (µm) nominal and 40 µm absolute rating shall be installed between the manufacturer’s plant system and the container to be filled for delivery.

3.4 Qualitative. The propellant shall be a clear, single-phased liquid when examined visually by transmitted light in accordance with (IAW) 4.3.1. Type NTO shall be red-brown in color; type MON compositions shall be green in color.

3.5 Leakage. Shipping containers shall not leak when tested IAW 4.4.

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TABLE I. ***Chemical and physical properties.***

Composition	NTO (Red-brown)	MON-1 (Green)	MON-3 (Green)	MON-10 (Green)	MON-15 (Green)	MON-25 (Green)	Test Paragraph
NO content \1 wt.%, min wt.%, max.	\2	1.0 0.6	3.0 2.5	11.0 10.0	16.0 15.0	26.0 25.0	4.3.2
N_2O_4 assay \3 wt.%, min.	99.5	---	---	---	---	---	4.3.3
N_2O_4 + NO wt.%, min.	___	99.5	99.5	99.5	99.5	99.5	4.3.3
Water equivalent wt.%, max.	0.17	0.17	0.17	0.17	0.17	0.17	4.3.4
Chloride \4 wt.%, max.	0.040	0.040	0.040	0.040	0.040	0.040	4.3.5
Non-volatile residue mg/L, max	---	10.0	10.0	10.0	10.0	10.0	4.3.6
Iron mg/kg (ppmw), max.	---	0.5	0.5	0.5	0.5	0.5	4.3.7
Particulate mg/L, max.	10.0	10.0	10.0	10.0	10.0	10.0	4.3.8

NOTES:

1 Nitric oxide
2 The NO content shall be limited to that which does not change the specified red-brown color of the propellant (3.4).
3 Dinitrogen tetroxide
4 The chloride test need not be performed on any propellant type if the material was manufactured by the ammonia oxidation process and is explicitly specified in the contract.

4. VERIFICATION

4.1 Classification of inspections. The inspections shall be classified as conformance inspections (see 4.2).

4.2 Conformance inspection. The conformance inspection shall consist of the following:

a. Individual tests (see 4.2.1)

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b. Sampling tests (see 4.2.2)

4.2.1 Individual tests. The propellant shall be subjected to the following tests as described under 4.3:

a. Examination of product (see 4.3.1)

b. Leak test (see 4.4)

4.2.2 Sampling tests. The propellant shall be selected IAW 4.2.2.1 and subjected to the tests indicated in Table I as described under 4.3.

4.2.2.1 Sampling plan. Unless otherwise specified (see 6.2), each filled shipping container shall be considered a lot and shall be sampled.

4.2.2.1.1 Sample. A sample consists of not less than 800 milliliters (mL) of propellant for all tests specified in 4.2.2. Unless otherwise specified, - conformance tests shall be performed on each sample of propellant taken directly from the shipping containers (other than the container loading filter specified in paragraph 3.3, product sampled shall not be exposed to additional filtration prior to sample analysis). When required, the sample shall be forwarded to a laboratory designated by the procuring activity for subjection to the - conformance tests specified herein.

4.2.2.1.1.1 Sampler. The sampler(s) shall be capable of accepting a minimum of 400 mL of product and constructed of 300 series stainless steel (SS). The top and bottom valves and any outlets or welds shall also be of 300 series stainless steel. Samplers shall be pre-cleaned in a manner such as to preclude contamination of propellant samples.

4.2.2.1.2 Method of taking sample. Samples taken from the shipping containers shall be representative of the entire contents of the container. The container's dip tube must be completely purged of product prior to taking the sample (approximately (~) 2.0 liters (L)).

4.2.3 Rejection. When any sample of the propellant tested IAW 4.3 fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected.

4.3 Test methods.

4.3.1 Examination of product. The propellant sample shall be visually examined to ensure that the material conforms to 3.4. The examination shall be conducted after a representative aliquot (~ 100 mL) of the sample has been transferred to a clear glass vessel (graduated cylinder or large test tube), cool in an ice-bath for 10 minutes (min), and wiped dry. Examine visually by transmitted light from a light bulb or sunlight. The presence of visible sediment in the sample shall be cause for rejection.

4.3.2 NO assay. The NO content of MON type samples shall be determined by one of the methods specified in Appendix A. In the event of a dispute, the method in A.2.1 shall be the referee.

4.3.3 N_2O_4 assay. The dinitrogen tetroxide content of the propellant shall be determined by the method described in Appendix B.

4.3.4 Water equivalent. The water equivalent content shall be determined by one of the methods specified in Appendix C. In the event of a dispute the method in C.2.1 shall be the referee.

4.3.5 Chloride. The chloride (Cl^-) content of the propellant shall be determined by one of the methods specified in Appendix D. In the event of a dispute the procedure specified in D.2.1 shall be the referee.

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4.3.6 Non-volatile residue (NVR). The NVR of the propellant shall be determined in duplicate by the method described in Appendix E.

4.3.7 Iron. The iron content of the propellant shall be determined by one of the sample preparations and methods described in Appendix F. In the event of a dispute, the sample preparation in F.2.1.1 and the method specified in F.2.2.1 shall be referees.

4.3.8 Particulate. The particulate matter content of the propellant shall be determined by the method described in Appendix G.

4.4 Leakage. Each shipping container will be checked for leaks at all potential sources for leaks such as valves and fittings. Leaks shall be tested for by utilizing a toxic vapor detector with a lower detection limit for NO_2 of at least 1.0 mg/L (ppmv). The detector wand shall be placed ½ - 1 inch from each potential leak source for at least 3-5 seconds. Any positive reading on the toxic vapor detector constitutes a leak.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The propellant described by this specification is intended for use as an oxidizer in rocket engines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and grade (if applicable) of propellant required (see 1.2).

c. Quantity by weight in pounds (avoirdupois).

d. Method of shipment, type, and capacity of containers.

c. When test methods are other than those specified (see 3.1).

d. When a different sampling plan is required (see 4.2).

6.3 Safety Data Sheets. Contracting officers will identify those activities that require copies of completed Safety Data Sheets (SDS) prepared in accordance with the 29CFR1910.1200.

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6.4 Subject term (key word) listing.

MON

Nitric oxide

Oxidizer

Rocket engine

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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NO ASSAY

A.1 SCOPE

A.1.1 Scope. This Appendix describes two methods for the determination of nitric oxide content in the MON type propellant samples. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 METHODS

A.2.1 Oxidation.

A.2.1.1 Sample preparation. Evacuate and weigh to the nearest 0.001 grams (g) a sample bomb of suitable capacity (≥ 60 mL) capacity, fitted with a needle valve. Support the sampler in a vertical position on a ring stand in a hood. Connect the sample bomb to the sampler and provide for evacuating the connecting line. Evacuate the connecting line, then, open the sample bomb and sampler valves. Transfer an aliquot of the sample of ≥ 40 g (bomb capacity dependent). Close all valves and then evacuate the connecting line. Disconnect the sample bomb and flush the exposed opening of the valve with nitrogen (N_2) gas conforming to MIL-PRF-27401, until residual N_2O_4 disappears. Accurately reweigh the sample bomb to the nearest 0.001 g, obtaining the sample weight by difference. If sample weight is below 40 g, another sample shall be taken.

A.2.1.2 Procedure. Attach a rotameter with SS tubing to the reducing valve of a cylinder of oxygen (O_2) gas conforming to MIL-PRF-25508. Connect a length of 1.6 millimeter (mm) ($^1/_{16}$ inch (in)) SS tubing to the exit end of the rotameter. Open the cylinder valve and purge residual air from the connecting lines and rotameter. Immediately attach the sample bomb to the other end of the 1.6 mm tubing and, using the reducing valve, adjust the O_2 pressure to 30 pounds per square inch (psi). Close the cylinder valve. Place the sample bomb in an ice bath and maintain the temperature at ~ 0°C (32°F). After the sample has been allowed to cool for 10 minutes (min), permit O_2 to flow into the sample bomb by first opening the cylinder valve and then the bomb valve. Occasionally shake the bomb to hasten absorption and oxidation. Lack of O_2 flow through the rotameter indicates completion of reaction. Allow the sample bomb to remain in the ice bath for 15 min, with valves open after reaction has been completed. Disconnect the sample bomb, remove from the bath, warm to ambient temperature and wipe dry. Place on a balance and record the weight to the nearest 0.001 g.

A.2.1.3 Calculation.

$$x = B - A$$

$$y = C - B$$

$$NO\ wt\% = \frac{\left[y\ - 0.003857\left(z - \frac{x + y}{1.49}\right)\right] 187.5}{x} - 0.15$$

where,

x = Sample weight, g

y = O_2 weight, g

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z = Volume of sample bomb, mL. Determined by difference in weight from evacuated and water filled

A = Weight of empty sample bomb, g

B = Weight of sample bomb plus sample, g

C = Weight of sample bomb plus sample and O_2, g

A.2.2 Spectrophotometric (alternate).

A.2.2.1 Preparation. Clamp the sampler specified in A.2.1 to a stand as shown in Figure A-1. Attach a 1.6 mm ($^1/_{16}$ in) ID length of polytetrafluoroethylene (PTFE) tubing to the outlet valve. Place another length of 1.6 mm ($^1/_{16}$ in) ID PTFE tubing in a beaker containing a strong sodium hydroxide (NaOH) solution. Set this apparatus up in a fume hood in which low pressure gaseous N_2, conforming to MIL-PRF-27401, is available for purging excess N_2O_4 from the tubing. The flow-through spectrophotometric cell depicted shall have a 2.00 ± 0.02 mm sample path length equipped with a coolant jacket for sample temperature control constructed of material compatible with N_2O_4.and must fit in the instrument cell compartment. Figures A-2 and A-3 show suggested designs. Maintain the temperature of the flow-through cell at 0 ± 1°C (32 ± 1.8°F) by connecting the cell to a circulating ice-water glycol (20:1) bath located at the apparatus in the fume hood and/or at the spectrophotometer. Maintain a purge of low pressure gaseous N_2 in the spectrophotometer cell compartment or on the cell assembly to prevent condensation of moisture on the cell windows when the cell is in the spectrophotometer.

A.2.2.2 Procedure. Clamp a cylinder containing red-brown NTO in the apparatus according to Figure A-1. Connect the spectrophotometric cell to the cylinder by means of 1.6 mm ($^1/_{16}$ in) ID PTFE tubing. Purge the cell with dry N_2 for several minutes. Close the inlet and outlet valves. Connect the cold bath to the cell and precool by pumping coolant for several minutes. Open the lower valve on the N_2O_4 cylinder slightly and control the sample flow with the cell valves. Pass several cell volumes (at least 10 mL) of sample through the cell. Make sure the cell is free of bubbles, then close the cell outlet and inlet valves and the N_2O_4 cylinder valve, in that order. If possible, purge the cell with dry N_2 before cooling, using several cell volumes of sample, and then checking for the absence of bubbles. Place the cell in the spectrophotometer cell compartment and connect to an ice-water-glycol bath through suitable tubing. Allow the coolant to circulate for ~ 5 min. Set the absorbance to zero (100%T) at a wave length of 700 nm. Remove the cell to the apparatus (Figure A-1) in the hood and expel the N_2O_4 into the scrubber with a low pressure stream of N_2. Repeat the above procedure substituting the sampler. The temperature of the sampler should be 25 ± 2°C (77 ± 3.6°F). Measure the absorbance of the sample at 0 ± 1°C (32 ± 1.8°F) and a wavelength of 700 nm. Determine the NO percent by weight from a calibration curve prepared according to A.2.2.3.

A.2.2.3 Calibration curve. A calibration curve shall be prepared as follows: Weigh accurately 58.164 g of American Chemical Society (ACS) reagent grade nickel nitrate hexahydrate [$Ni(NO_3)_2 \cdot 6H_2O$] or equivalent, to make 100 mL of a 2.0 molar (M) stock solution. In a similar manner weigh out 26.008 g of sodium azide (NaN_3) to make 100 mL of a 4 M stock solution. Filter or decant the latter after allowing it to stand overnight. Standard A, equivalent to 0.5 wt% NO, is prepared by placing exactly 1.00 mL of the 2 M Ni^{2+} solution in a 50 mL beaker. Add 4.00 mL of the 4 M NaN_3 solution and 15.0 mL purified water (defined throughout this document as water free of interfering substances and 18 Mohm or greater resistivity, unless otherwise specified) and mix. Standard B, equivalent to 0.86 wt% NO, is prepared by mixing 1.00 mL of the 2 M Ni^{2+} solution with 4.0 mL of the 4 M NaN_3 solution, and 8.34 mL of purified water in a second 50 mL beaker. Fill a 5.0 mm absorption cell with purified water as a blank. Set the spectrophotometer at 700 nm. Set the blank at 100%T (0 absorbance). Determine the absorbance of standards A and B in the same 5.0 mm cell at 700 nm. Plot the absorbance of Standards A and B versus

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the equivalent wt% NO (x-axis) and draw a straight line through the origin (0 wt% NO). The mixed synthetic standard solutions are stable for 48 hours. The sodium azide stock solution is stable for 30 days in the dark. The nickel nitrate stock solution is stable indefinitely.

A.2.2.4 Reagents:

a. Purified water as defined in A.2.2.3

b. Nickel nitrate hexahydrate, ACS reagent grade or equivalent

. c. Sodium azide, ACS reagent grade or equivalent

d. Gaseous nitrogen conforming to MIL-PRF-27401

e. Gaseous oxygen conforming to MIL-PRF-25508

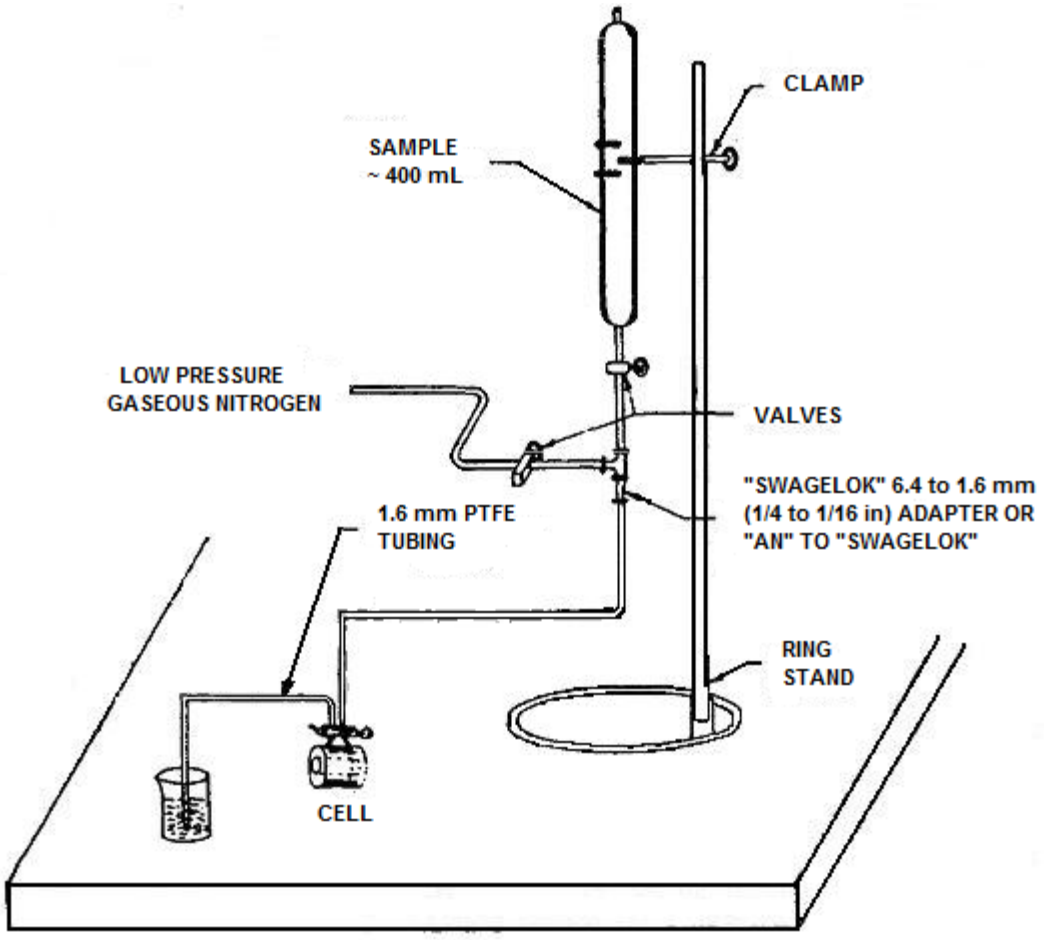


FIGURE A-1. ***Nitric oxide sampler apparatus***

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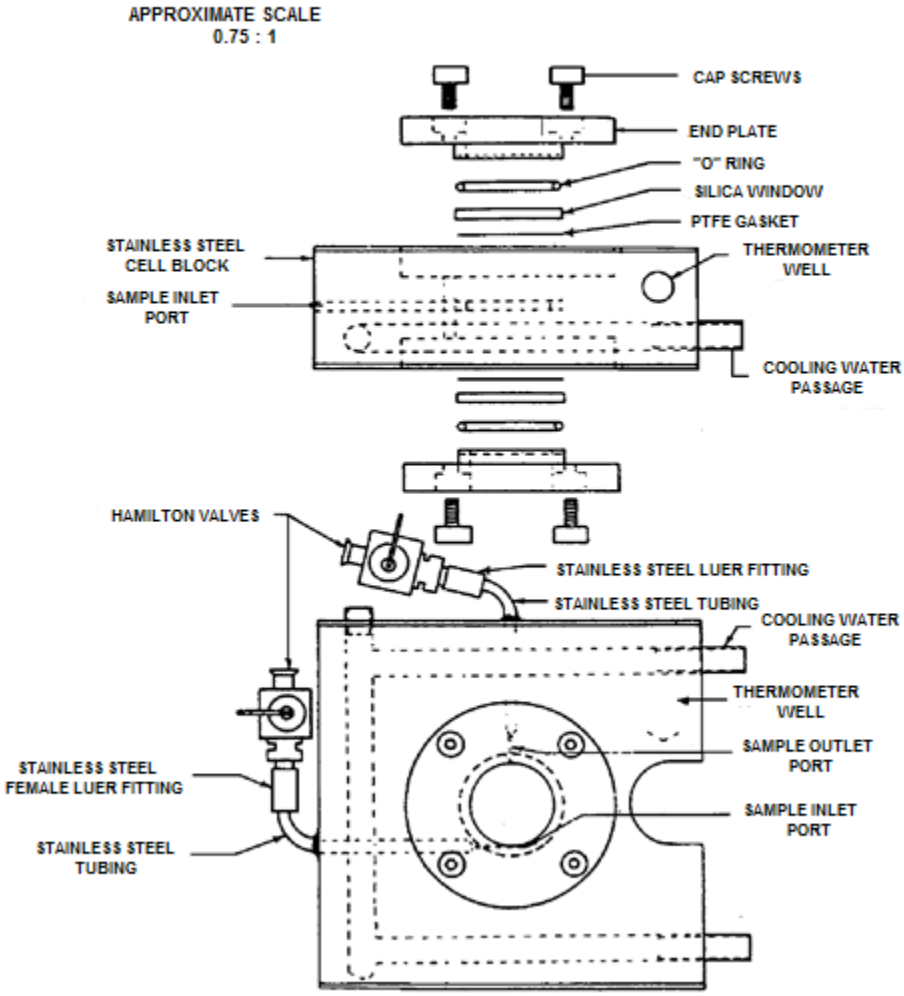


FIGURE A-2. ***Stainless steel absorption cell***

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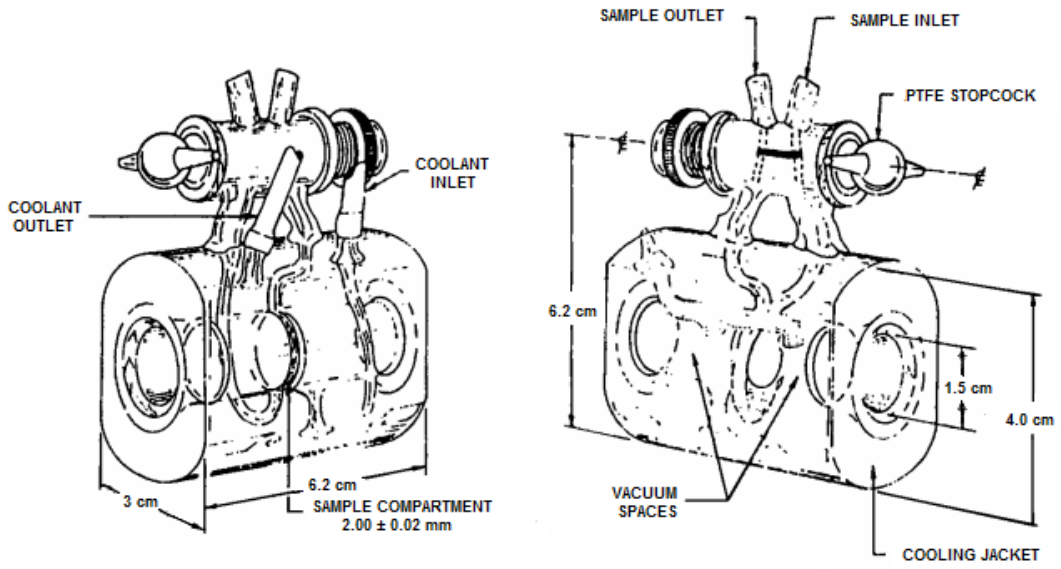


FIGURE A-3. ***Glass absorption cell.***

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N_2O_4 ASSAY

B.1 SCOPE

B.1.1 Scope. This Appendix describes a method for the determination of the dinitrogen tetroxide content of the propellant. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 METHOD

B.2.1 Sample preparation.

B.2.1.1 Oxidation state. Prior to the weighing and sealing of the assay sample, comply with one of the following requirements.

B.2.1.1.1 NTO composition. NTO shall be assayed without further chemical treatment. Proceed to B.2.1.2 and then to B.2.1.3.

B.2.1.1.2 MON composition. The N_2O_4 assay of MON material shall be performed without further chemical treatment on the oxidized sample specified in A.2.1. Proceed to B.2.1.2 and B.2.1.3. If the NO content of the MON material was determined by the alternate method (spectrophotometric), then, a 10 − 15 mL sample shall be oxidized as specified in A.2.1 before proceeding to B.2.1.2 and B.2.1.3 below.

B.2.1.2 Sample sealing. Cool the sample to ~ 0°C (32°F) with crushed ice. Introduce the cooled sample into a clean, weighed (2 mL) glass ampoule directly from the sampler through a 1.6 mm ($^1/_{16}$ in) SS tube. Insert the ampoule tip into an adequate size polyvinylchloride (PVC) tube, ~ 15 cm (6 in.) long. Freeze the contents of the ampoule by placing in a tray of crushed dry-ice or by some other appropriate method. Lift the ampoule containing frozen N_2O_4, wrap it with a piece of soft paper, and remove the PVC tube. Seal the ampoule in the hot flame of a methane/oxygen (CH_4/O_2) or hydrogen/oxygen (H_2/O_2) torch by rotating at an angle. Test for leakage. Wash, dry, and weigh the ampoule. Sample sizes should average ~ 1.3 g when 0.5 N alkali is used in B.2.1.3. (Alternate procedures may be used provided sample integrity is not compromised).

B.2.1.3 Procedure. Place the sealed ampoule into a heavy-walled glass stoppered Pyrex bottle, or iodine flask, containing 100 mL of purified water plus 20 mL of 30% hydrogen peroxide (H_2O_2). Chill contents in ice water or some other appropriate method. Lubricate the stopper with stopcock grease and insert it into the ground glass neck of the bottle and/or flask. Break the sealed ampoule by shaking vigorously, keeping the stopper firmly in place. Continue shaking until all brown fumes disappear. Return the bottle or flask to the ice water to keep the contents cool. Shake the bottle from time to time over a 15 min period. Remove the stopper, rinse it off with purified water into the contents of the bottle or flask, and replace it with an air condenser. Remove this assembly from the ice water and place on a steam bath or hot plate at moderate heat setting (~ 120°C/250°F) for 45 min. Allow the contents to cool to ambient temperature. Rinse the air condenser carefully and remove it. Pulverize the broken glass of the ampoule with the flat end of a heavy glass rod. Rinse the rod with purified water into the bottle or flask. Add 3 – 5 drops of methyl red indicator and titrate with 0.5 N sodium hydroxide (NaOH) solution to the end point. Run a reagent blank in the same manner as the sample.

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B.2.1.4 Calculation. The percent N_2O_4 shall be calculated by one of the following formulas:

B.2.1.4.1 NTO

$$N_2O_4\ wt\% = \frac{[(a-b)N]4.6008}{W}$$

where,

a = Volume of NaOH used for sample, mL

b = Volume of NaOH used for blank, mL

N = Normality of NaOH

W = Weight of N_2O_4 in the ampoule, g

B.2.1.4.2 MON Composition

$$N_2O_4\ wt\% = 1.534\ (A\ - B)$$

where,

$$A\ = \frac{[(a-b)N]\ [3.00\ +0.016(\%NO)]}{W}$$

a = Volume of NaOH used for sample, mL

b = Volume of NaOH used for blank, mL

N = Normality of NaOH

W = Weight (g) of oxidized N_2O_4 in the ampoule

B = Wt% NO

B.2.1.5 Reagents. The following shall apply as test conditions of B.1:

(1) Purified water as defined in A.2.2.3.

(2) Sodium hydroxide, 0.5 N, standardized: Dissolve 20 g of ACS, or equivalent, reagent grade, low carbonate, NaOH in distilled water in a 1000 mL volumetric flask. Allow to cool and dilute to the mark with distilled water. Standardize this solution with standard potassium acid phthalate (KHP) to the phenolphthalein end point. Store the 0.5N NaOH in a polyethylene container and ensure the exclusion of carbon dioxide during storage. (May be commercially obtained).

(3) Potassium acid phthalate (Primary Standard Grade): Dry at 105°C for 2 hours. Weigh to ± 0.1 mg and dissolve in purified water.

(4) Phenolphthalein indicator solution: 1 g in 100 mL of anhydrous denatured ethyl alcohol

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(5) Hydrogen peroxide, ACS reagent grade, 30% by wt. or equivalent.

(6) Methyl Red indicator solution: 0.1 g in 100 mL of purified water.

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WATER EQUIVALENT

C.1 SCOPE

C.1.1 Scope. This Appendix describes two methods for the determination of the water equivalent content of the propellant. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

C.2 METHODS

C.2.1 Near infrared absorption method.

C.2.1.1 Preparation. Optimize the conditions of the spectrophotometer. Table C-I shows suggested parameters for initial setup. These parameters will vary depending on model of spectrophotometer. Set the temperature of a circulating low temperature bath to −6 ± 1°C (21.2 ± 1.8°F). Attach the cell to the cooling bath and start the circulation. Purge the sample compartment with N_2 gas conforming to MIL-PRF-27401 so that the cells do not become coated with moisture. Throughout the procedure, minimize the exposure of the oxidized N_2O_4 to the atmosphere.

TABLE C-I. *Conditions for the spectrophotometer*.

Method	Scan/Manual
01 Ord/Absc Mode	Abs/WL
02 Slit/Gain/Sens	2.00 nm/ /2
03 Scan Speed	240 nm/min
04 Response	0.5 S
05 Lamp/Det	319.2 / 860.8 nm
06 Cycles/Time	1 / 0.05 min
07 Peak Threshold	0.01 A
08 Recorder	SERIAL / DASH 1
09 Ord Min/Max	.0000 / 0.500
10 Absc Min/Max	1300.0 / 1650.0
11 Abscissa Format	20 nm/cm
12 Printer	/ SCALE /

C.2.1.2 Procedure. Transfer some of the oxidized N_2O_4 to a small cooled Erlenmeyer flask. Load the cell from the flask using a cooled 2 mL glass syringe, and place in the sample beam of the spectrophotometer. Place an empty cell in the reference beam. Set the spectrophotometer to1300 nm, then, zero the instrument. Push the RUN/STOP button to start the scan. The absorption of the HNO_3 band pass has to be measured from the baseline which is found by connecting the minimal points at 1430 nm and 1615 nm. The band pass of atmospherical (free) water can be found by entering 1404 PEAK FIND. Record the absorbance.

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C.2.1.3 Calculations. The water equivalent shall be calculated by one of the following formulas:

NTO composition.

$$Water\ equivalent\ wt\% = \frac{A_{HNO_3}}{hbf} + \frac{A_{H_2O}}{ab}$$

MON Composition.

$$Water\ equivalent\ wt\% = \left[\frac{A_{HNO_3}}{hbf} + \frac{A_{H_2O}}{ab}\right]\left[\frac{100 + C(\%\ NO)}{100}\right]$$

where,

A_{HNO_3} = Absorption of HNO_3

A_{H_2O} = Absorption of free water

h = Absorption coefficient of HNO_3 = 0.178

b = Thickness of sample (cm)

f = Factor for conversion of HNO_3 to H_2O = 7

a = Absorption coefficient of free water = 0.55

C = Calculation coefficient (NO to N_2O_4) = 0.534

C.2.2 Gas chromatographic method..

C.2.2.1 Analytical apparatus. Assemble the analytical apparatus as shown in Figure C-1. Connect one side of a 2 microliter (µL) SS liquid inlet sampling valve to a multi-port, multifunctional switching valve. Connect the other side of the same 2 µL valve to a 30.5 cm x 0.95 cm (12 in x $^3/_8$ in) Vycor glass tube filled with copper filings and loosely plugged with copper turnings or quartz wool at each end. The copper filings should be of sufficient surface area to reduce 5 or 6 injections of N_2O_4 without peak deterioration but large enough in size to not restrict gas flow. These may be produced by a rough file on copper metal. Place the copper packed reactor in a split tube furnace capable of maintaining a constant temperature between 550 and 600°C (1022 and 1112°F). Connect the other end of the copper-packed reactor to the separation column of the gas chromatograph (GC) through 1.6 mm ($^1/_{16}$ in) SS tubing. This is the sample inlet for the separation. The distance between the copper reactor and GC must be 5 cm (2 in) or less and insulated with glass wool with an aluminum foil over-wrap. Do not heat interconnecting lines; insulation may be used where necessary. The inlet system should also contain a septum inlet or sample loop or loops of known volume (in the range of 10 to 100 µL) connected to the switching valve. This is used for calibration of the GC. The column of the GC should be of sufficient length of 3.2 mm ($^1/_8$ in) tubing and contain a support material to adequately resolve water when N_2 is used as the carrier gas (diatomaceous earth supports or other water absorbing supports shall not be used as support materials). The detector should have a minimum sensitivity of 0.25 millivolt (mV) per microgram (µg) of water. The reference and sample flow rates should be matched within 5%. Optimize the column and detector temperatures to produce sharp peaks with adequate resolution. To regenerate the reactor, engage the switching valve to the hydrogen (H_2) gas source (pre-purified grade, 99.95%, 20 ppm O_2 max.) and isolate the GC. When the copper reactor is disconnected from the GC for isolation,

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turn off the detector to prevent air contact. Regeneration is accomplished by alternating small amounts of hydrogen with the carrier gas flow (N_2 conforming to MIL-PRF-27401 or helium (He) conforming to MIL-PRF-27407) into the apparatus until no more water can be detected at the outlet or when black copper oxide has returned to bright metallic copper. Dead volume must be kept to a minimum throughout the system by using 1.6 mm ($^1/_{16}$ in) OD tubing of the shortest length possible and filling voids in furnace connections with copper wool.

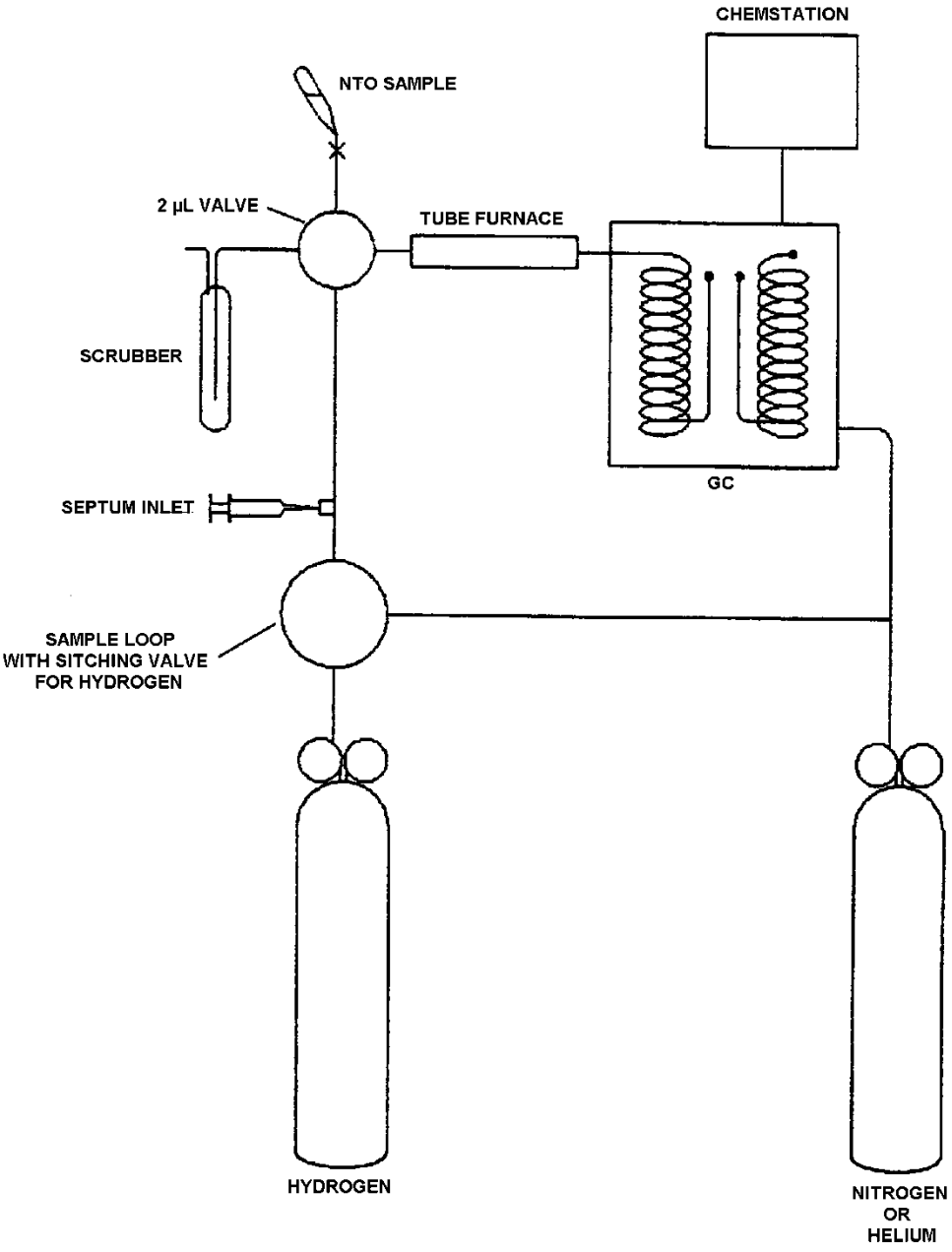


FIGURE C-1. ***Water equivalent apparatus, schematic.***

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C.2.2.2 Procedure. The NTO sample specified in B.2.1.1 shall be connected to the 2 µL liquid valve shown in Figure C-1 by means of a 3.2 mm ($^1/_8$ in) SS tube. The MON material shall be oxidized according to A.2.1 prior to connecting it to the 2 µL valve. Excess oxygen in the vessel head space shall be released into the NTO scrubber and the vessel pressurized to 30 – 60 psi with dry N_2 or He. The apparatus specified in Figure C-1 shall be in operational mode ready to analyze a 2 µL aliquot. When the vessel containing the oxidized NTO is connected to the 1.6 mm ($^1/_{16}$ in) line leading to the 2 µL valve, the vessel should at no time be held directly vertical with the valve downward. Connect and clamp the vessel at a slight angle from vertical. It should also not be agitated prior to connection. These precautions are necessary to prevent any particulate from entering and plugging the 2 µL valve. Allow the NTO to flow through the 2 µL sample valve to the NTO scrubber. When a steady stream of liquid flows introduce the aliquot and stop the NTO flow. Reject the first run after regeneration as it will be high due to removal of dead volume H_2. If a manual valve is used switching must be rapid and consistent with a low NTO flow to avoid valve leakage. Repeat the analysis two more times. Determine the areas or peak heights. When the peak heights or areas of subsequent injections after the first rejected injection begin to decrease regenerate the copper in the reaction tube. Repeatability should be within 0.02% water equivalents. If repeated injections are out of this range a dead volume check should be performed by checking all connections for proper filling of copper wool and the injection valve should be evaluated for leaking.

C.2.2.3 Calculation. Average the peak heights or areas and determine the micrograms (µg) of water from a previously prepared calibration graph (C.2.2.4). Calculate the water equivalent content using the following formula:

$$Water\ equivalent\ wt\% = \frac{\mu g\ H_2O}{29}$$

The peak heights or areas of the duplicates must be reproducible within 2%. If they are not, premature sample vaporization and/or a contaminated valve may be responsible for variations.

C.2.2.4 Calibration. A calibration curve shall be prepared as follows: Using a pressure-lok gas syringe having a 0 – 50 µL range, inject a 5 µL sample of H_2 (at ambient temperature) into the partially oxidized copper reactor and record the peak height and the area for the resultant water peak. Repeat this using 10, 25, and 50 µL of H_2 gas. Calibration may also be performed by using a sample loop of known volume connected to a switching valve. The volume may be determined by the difference in weight when filling the loop with a solvent of known density. Record peak heights and areas for each sample. The calculation to convert µL of hydrogen for each sample to µL of water is as follows:

$$\mu g\ H_2O = \frac{18\ [a(P)(273)]}{(22.4)(760)T} = \frac{0.2887\ (aP)}{T}$$

where,

a = Sample volume, µL

P = Ambient barometric pressure, mmHg

T = Ambient temperature, °K

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Construct a calibration graph of peak height or area versus µg H_2O and draw a smooth line through the origin. A single point calibration may be used using one sample loop to confirm the calibration in subsequent analyses. This evaluates the detector or column deterioration while preventing the leaking of syringe ports or connections to sample loops deteriorated from multiple removal. The single point calibration should be performed before and after each set of samples.

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CHLORIDE

D.1 SCOPE

D.1.1 Scope. This Appendix describes two methods for the determination of the chloride content of the propellant. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

D.2 METHODS

D.2.1 Silver nitrate titration.

D.2.1.1 Sample preparation. Place a 250 mL beaker on a magnetic stirrer in a laboratory hood. Add to the beaker a PTFE stirring bar and 150 mL 1 M NaOH solution. Activate stirrer. Weigh a sampler containing NTO or oxidized MON material (A.2.1) to 0.1 g. A sample vessel that holds approximately 100 mL is best for weighing accuracy. Attach a PTFE tube to the sampler. Slowly add 6 to 8 mL (9 to 12 g) of sample via the tube below the surface of the caustic. The rate should be low enough so that no red vapors form from breaking bubbles which will reduce the amount of sample introduced. Remove the tube and determine the sample weight by difference. Slowly add with stirring an amount of urea equal to one-half (0.5) the sample weight in grams. Care should be taken to prevent sample loss due to bubbling. Slowly add with stirring 1:1 HNO_3 until gas evolution ceases or the pH stabilizes between 2.5 to 3.0. The pH of the solution should be checked often while adding the acid to make sure that a pH less than 2.5 is not produced. Allow the solution to stand 15 min in preparation for titration.

D.2.1.2 Procedure. Fill a syringe pump, adjusted to deliver a constant volume of titrant per unit time (in the range of 0.5 to 0.7 mL/min) with 0.05 M silver nitrate ($AgNO_3$) solution. Connect the pump to the prepared sample with a PTFE tube. Place a chloride or silver standard ion electrode and a double-junction reference electrode in the sample solution. The electrode should be connected to a millivolt meter and recorder. With sample stirring, turn on meter, recorder, and pump. Chart speed should be ~ 2.5 cm/min (1 in/min). Mark the point where the syringe pump delivery begins. As soon as the S shape curve is complete, turn off the pump and recorder. Determine the inflection point by drawing tangents to the curve, dropping a perpendicular from the upper to lower tangent, and constructing a perpendicular bisector. Measure the exact titrant delivery time, such as 5.75 min. Run a blank in the same manner. Delivery time of titrant in minutes multiplied by delivery rate in mL/min equals titrant volume in mL. Alternately the S shape may be generated by incremental additions of the $AgNO_3$ solution with a 25 mL burette and detection of millivolts from the millivolt meter. The millivolts vs milliliters shall be plotted during $AgNO_3$ addition to determine when the S curve is approaching. Increments should be 1 mL or less to the inflection point and 0.25 or less at the rise to the inflection point, and returning to 1 mL increments after the rise. An automatic titrator may also be substituted for the syringe pump provided the detection uses the same electrodes and the end point is identical to the plotted S curve. The $AgNO_3$ may require dilution for automatic titrators.

D.2.1.3 Calculation. Calculate the chloride content of the propellant using the following formula:

$$Chloride\ wt\% = \frac{(mL\ AgNO_3 - mL\ blank)(M)(35.45\ \mathrm{g/mol})100}{W}$$

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where:

M = Molarity of $AgNO_3$ titrant (moles/L)

W = Sample weight in mg

D.2.1.4 Calibration. The $AgNO_3$ molarity (M) is obtained by titrating, in triplicate, a standard sodium chloride solution by the procedure in D.2.1.2.

D.2.2 Colorimetric method.

D.2.2.1 Procedure. Weigh the sample bomb containing the oxidized sample prepared in A.2.1 to the nearest 0.10 g. Slowly bubble 1 to 50 g of N_2O_4 through 40 to 50 mL of purified water contained in each of two 250 mL gas scrubbers, connected in series with polyvinyl tubing. The sample must be scaled to contain less than 0.050 mg Cl^-. Combine the two solutions in a 250 mL beaker. Prepare a reagent blank with 50 mL of purified water in a second beaker. Reweigh the sample bomb. Add 2 to 5 drops of 2,4-dinitrophenol (0.1% by wt. in alcohol and water) to each of the 250 mL beakers and neutralize each with concentrated ACS reagent grade ammonia to the yellow end-point. Adjust the volume in the sample beaker to contain one gram of N_2O_4 per mL of solution by evaporating on a hot plate. Into three other 25 mL volumetric flasks add 0.010, 0.020, and 0.040 mg Cl^- of the dilute standard Cl^- solution, using a microburet. Add purified water to all flasks so that they contain ~ 20 mL of liquid. Add exactly 2.0 mL of a 0.25 M iron (III) ammonium sulfate [$FeNH_4(SO_4)_2 \bullet 12H_2O$] solution to the flasks and mix. Add exactly 1.0 mL of a saturated mercury (II) thiocyanate [$Hg(SCN)_2$] solution, bring up to the mark with purified water and mix. Place all flasks in darkness for 15 to 30 min. Measure the absorbance of the sample and the standards in a 5.00 cm cell at 460 nm, after setting the blank in a 5.00 cm cell at zero absorbance (100%T). Plot the absorbance of the standards versus mg Cl^- and draw a curve through the origin.

D.2.2.2 Calculation. The weight percent chloride shall be calculated as follows:

$$Chloride\ wt\% = \frac{(mg\ Cl^- from\ curve)(F)(100)}{mg\ N_2O_4}$$

where,

F = Aliquot dilution factor

D.2.3 Alternate analytical methods. The chloride content of the propellant may also be determined by other scientifically accepted methods, such as ion chromatographic techniques, as long as the instruments are calibrated with certified reference standards and are able to quantitate Cl^- at the concentrations required by this specification.

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NON-VOLATILE RESIDUE (NVR)

E.1 SCOPE

E.1.1 Scope. This Appendix describes a method for the determination of non-volatile residue of the propellant. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

E.2 METHOD

E.2.1. Hot plate. This method involves the gradual evaporation of an N_2O_4 sample on an explosion proof hot plate in a fume hood. The difference in weight before and after evaporation is calculated as NVR.

E.2.1.1 Procedure. Place two platinum evaporation dishes per sample in an oven at 110°C (230°F) for 30 min. Remove evaporation dishes from the oven, cover with a watch glass and allow to cool next to the balance for 30 min. Weigh each clean, dry evaporating dish to the nearest 0.01 mg and record the weight. Place the evaporating dishes on a hot plate in an exhaust hood and transfer 50 mL of the N_2O_4 to each dish. Set the hot plate to a temperature of 150 – 200°C and allow the propellant to evaporate to dryness. Remove the evaporating dishes from the explosion-proof hot plate and oven dry for one hour. Remove dishes from the oven, cover with a watch glass and allow to cool next the balance for 30 minutes. Weigh each dish to the nearest 0.01 mg. Repeat oven drying until constant weight is attained.

E.2.1.2 Calculations. Calculate the nonvolatile residue of the propellant as follows:

$$NVR\ mg/L = \frac{Wt_S - Wt_E}{V}$$

where,

Wt_S = Weight of the sample evaporating dish after drying, mg

Wt_E = Weight of the empty evaporating dish, mg

V = Volume of sample used, L

E.2.1.3 Equipment.

a. Hot plate, explosion proof

b. High efficiency fume hood

c. Five decimal place analytical balance or better

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IRON

F.1 SCOPE

F.1.1 Scope. This Appendix describes two sample preparation procedures and three methods for the determination of the iron content of the propellant. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

F.2 METHODS

F.2.1 Sample preparation

F.2.1.1 Referee. Using a 50 mL graduated cylinder, transfer two 50 mL aliquots of the propellant into two evaporation dishes. Use larger sample sizes when less iron is expected. Place the evaporating dishes on a hot plate and set to the low heat range. Slowly evaporate the sample (avoid vigorous boiling) to near dryness and residue becomes colorless (~ 1 – 2 mL). Remove the sample from the hot plate, add 1 mL concentrated H_2SO_4, swirl to mix and allow to cool for ~ 5 min. (**NOTE:** A straw color is an indication of N_2O_4 remaining). Transfer the contents of the vessel to a 25 mL volumetric flask. Rinse the vessel with purified water and transfer the rinse to the 25 mL volumetric flask. Dilute the sample in the flask to the mark with purified water. Stopper the flask and shake to ensure complete mixing.

F.2.1.2 Alternate. Rinse the propellant residue contained in the evaporation dishes from 4.3.6 with 1 mL of concentrated H_2SO_4 (suitable for trace metal analysis) and transfer the contents quantitatively to a 25 mL volumetric flask. Dilute the sample in the flask to the mark with purified water. Stopper the flask and shake to ensure complete mixing.

NOTE: Nitric acid may be substituted for sulfuric acid in the sample preparation procedures under F.2.1 provided standards and samples are matrix matched.

F.2.2 Sample analysis

F.2.2.1 Inductively Coupled Argon Plasma-Atomic Emission Spectroscopy.

F.2.2.1.2 Procedure. For each instrument interference effects must be evaluated, corrected as applicable (ASTM D1976), and instrument performance validated. The argon gas supply must have a purity of 99.99% or better. Choose the iron wavelength recommended by the instrument manufacturer or that which provides the highest sensitivity (most commonly used is 259.940 nm).

F.2.2.1.3 Calibration. Prepare an Fe standard in a 4% H_2SO_4 acidified purified water solution to bracket the expected concentration range of the samples by combining an appropriate aliquot of a certified Fe stock standard. Prepare a mid-range check standard, from a different stock standard source than that used to calibrate the instrument to verify the calibration. Determine the concentration from the calibration curve prepared from the iron standard using an acid matrix blank as the zero point, and record the iron concentration of the sample.

F.2.2.1.4. Quality control. The instrumental drift shall be checked using the acid matrix blank and calibration standard at the beginning of each run, at least every ten readings during the run, and at the end of the run. The blank shall read ≤ the detection limit and the check standard shall fall within ±10% of the standard value. If the recovery of the analyte falls outside of the required control limits, the source of the problem must be identified and resolved before continuing analyses.

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F.2.2.1.5 Calculations. The iron content of the MON type propellant shall be calculated as follows:

$$Fe\ ppmw\ = \frac{(Fe_{ppmv})(V_f)}{V_i(D)}$$

where,

Fe_{ppmv} = Iron content of the sample solution, from the calibration curve

V_i = Volume of N_2O_4 used, mL

V_f = Final dilution of residue, mL

D = Density on N_2O_4 (1.45 g/mL)

F.2.2.2 Atomic absorption spectrophotometer.

F.2.2.2.1 Procedure. Determine the absorption of the sample on a suitable atomic absorption spectrophotometer at an iron wavelength closest to 2480 Å after setting the zero absorption with the acid matrix blank.

F.2.2.2.2 Calibration. Prepare a minimum of three Fe standards, and a mid-range check standard in the same manner as described in F.2.2.1.3. Determine the iron concentration in the sample from the calibration curve using the acid matrix blank as the zero point and record the results.

F.2.2.2.3 Calculations. Calculate the iron content of the propellant using the formula in F.2.2.1.5.

F.2.2.3 Colorimetric.

F.2.2.3.1 Procedure. ASTM E394 shall be followed with the following exceptions:

a. Add 5 mL of the ammonium acetate-acetic acid buffer solution to each empty 100 mL volumetric flasks (for standards and samples) and set aside.

b. Following the transfer of the contents in F.2.1.1 or F.2.1.2 alternatively, carefully add 50 mL of purified water. Mix thoroughly.

c. Add 1 mL of concentrated acid (H_2SO_4 or HNO_3 as applicable) and 50 mL purified water to at least four empty 150 mL evaporating dishes for the standards.

NOTE: Make sure solution is thoroughly mixed after each component addition.

d. Add the 2 mL of hydroxylamine hydrochloride and the 5 mL of 1,10-phenanthroline as described in the method.

e. Add enough sodium acetate (instead of the buffer called out in the method) to bring the pH of the mixture to between pH 3 and pH 4 while swirling (add ~ 20 mL initially and then continue adding drop wise until desired pH is reached).

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f. Transfer the solutions to their respective 100 mL volumetric flask from step a above, dilute to the mark, and mix thoroughly. Allow the flasks to set at room temperature in the dark for a minimum of 15 minutes.

g. Determine iron concentrations from the calibration curve prepared from the iron standards; the correlation coefficient shall be ≥ 0.990.

F.2.2.3.2 Calculations. Calculate the iron content of the propellant using the formula in F.2.2.1.5.

F.2.2.3.3 Reagents. The following reagents are in addition to those reagents listed in ASTM E394,

a. Sodium acetate solution. Dissolve 200 g $CH_3COONa•3H2O$ in 1 L volumetric flask with 700 mL with purified water and dilute to the mark. An equivalent solution may be obtained commercially.

b. Sulfuric or nitric acid, concentrated, suitable for trace metal analysis.

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PARTICULATE

G.1 SCOPE

G.1.1 Scope. This Appendix describes the method for the determination of particulate matter content of the propellant. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

G.2 METHOD

G.2.1 Particulate determination.

G.2.1.1 Preparation of membrane filter. Using forceps, place a 47 mm, PTFE membrane filter of nominal pore size (10 µm) in a clean Petri dish. Place the Petri dish, with its lid slightly ajar, in an oven of the static type at 90 ± 5°C (194 ± 9°F) and leave it for 30 minutes. Remove the Petri dish from the oven and place it near an analytical balance (must have a minimum precision standard deviation of 0.07mg). The Petri dish cover should be ajar but still protecting the membrane filter from contamination from the atmosphere. Allow 30 minutes for the membrane filter to come to equilibrium with the ambient air temperature and humidity. Remove the membrane filter from the Petri dish with forceps, handling it by the edge only, and place it on the center of the weighing pan. Weigh and return it to the Petri dish.

G.2.1.2 Sample. At minimum, a 400 mL aliquot of the sample should be passed through the filter assembly. The sample volume actually employed must be reported.

G.2.1.3 Procedure. Assemble the high pressure filter holder with the weighed membrane filter and position it so that the filter membrane is in a horizontal position. Pass a minimum of 400 mL of sample through the filter holder. Follow with dry nitrogen gas conforming to MIL-PRF-27401 to purge remaining N_2O_4. Carefully remove the membrane filter, and place it in a clean, covered Petri dish. Dry and reweigh the membrane filter taking care not to disturb the contaminant on its surface. The drying procedure described in G.2.1.1 should be followed in detail.

G.2.1.4 Calculation. Subtract the initial weight of the test membrane filter, W1, from the final weight, W2. Divide the resulting weight by the volume of sample filtered. Report the result to the nearest 0.01 mg/L and the actual sample volume used.

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CONCLUDING MATERIAL

Custodians
Navy – AS
Air Force – 68
DLA – PS

Preparing Activity
Air Force – 68
(Project: 9135-2017-001)

Review Activities
Air Force - 19

Civil Agency
NASA - NA

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